

**General Specification
for HVAC System**

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1. GENERAL

This document defines the design basis and criteria for the heating, ventilation and air conditioning (HVAC) system to be provided for Karabatan Power Plan, located at Karabatan, Atyrau region of western Kazakhstan, in Central Asia.

2. CODES, STANDARDS, REGULATIONS AND REFERENCES

2.1. APPLICABLE STANDARDS AND CODES

Unless otherwise specified, the latest edition of the following standards shall be used. Exceptions to the listed standards may be considered on case-by-case basis. Vendor shall submit any exceptions and deviations giving reference to clause no. of the referred standard for approval.

Vendor for his Scope of Supply shall be responsible that all aspects of the design, fabrication, inspection and testing conform to requirements of the specified standards.

The HVAC System design, installation, materials, testing and commissioning shall be in accordance with the following standards and codes:

- ASHRAE - American Society of Heating, Refrigerating and Air Conditioning Engineers;
- ASME - American Society of Mechanical Engineers;
- CIBSE - Chartered Institute of Building Services Engineers
- BSCP - British Standards Institute Codes of Practice
- IEC - International Electrotechnical Commission;
- ISO - International Standards Organization;
- NFPA - National Fire Protection Association;
- SMACNA - Sheet Metal and Air Conditioning Contractors National Association
- Standards, Codes or Regulation in use in Russian Federation and Kazakhstan (SNIP, GOST, etc).

Applicable Project Standards and Codes include, but are not limited to, the following Russian Federation publications:

- SNIP 41-01-2003 - Heating, Ventilation and Air Conditioning
- SNIP 23-01-99 - Building Climatology. Construction Norms and Regulations
- SNIP 2.04-05-91 - Heating, Ventilation and Air conditioning
- SNIP RK 2.04-01-2010 - Building Climatology
- SNIP RK 4.02-42-2006 - Heating, Ventilation and Air conditioning
- SNIP 2.04-01-85 - Interior Plumbing and Sewage in Building
- GOST 12.1.005-88 - Occupational Safety Standards. Sanitary requirements for working zone air
- SNIP 23-03-2003 - Noise Protection

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- SNIP 23-02-2003 - Thermal Protection of Building
- SNIP 21-01-97 - Fire Safety of Buildings and Structures
- VSN 21-77 - Instruction on designing heating and Ventilation Oil Processing and Petrochemical Enterprises
- NPB 105-03 - Determination of Explosion/Fire and Fire Hazard Categories of Rooms, Building and Outdoor Installations
- PB 09-540-03 - Explosion Safety Rules for Chemical Petrochemical or Oil Processing Plants

2.2. REFERENCE DOCUMENTS

- 01000-EM-SPC-003: General Site Information: Meteorological, Geological and Utilities Data
- Others to be specified by Contractor

2.3. DEVIATIONS FROM PROJECT AND INTERNATIONAL STANDARDS

Vendor shall quote for equipment as defined by the Work Requisitions. However, where he can offer equipment, which although deviating from dedicated project standards offers a significant technical and/or commercial advantage, he shall offer this equipment as an additional alternative for the Contractor's consideration. Any deviations from the requirements will be considered when specifically itemized within the quotation or listed separately as annex to the quotation, but will not be accepted unless agreed in writing by the Contractor.

All deviations thus accepted for the successful bid will be incorporated into order documentation. Subsequent to acceptance of an order, any further requests for deviations shall be routed through the Contractor's concession request system.

The lack of stated exceptions shall confirm full conformity with the specified requirements. Any conflict between the requirements of this specification and Supply Specifications and Data sheets I drawings, related standards and the order shall be referred to the Contractor by the Vendor for clarification. Vendor is prohibited to draw any conclusion himself in case of such conflicting requirements.

In the event of a conflict of requirements within the national or local laws and regulations, project standards and international standards the order of precedence shall be:

- a) National and/or local laws and regulations applicable at the location of the site
- b) International Standards as referenced in this General Specification
- c) Project specific catalogue data sheets from Licensor

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d) Project and Contractor standards

e) Manufacturer standards

It is the responsibility of the Vendor to contact the Contractor for resolution of any conflict and secure a written clarification from the Contractor.

3. FACILITIES TO BE PROVIDED WITH HVAC SYSTEM

HVAC Systems shall be provided for the following buildings of the Power Generation Area:

- Electrical & Distribution Substation Building
- Control Room Building
- Turbine Hall
- Pump House Room
- Clean Agent Room
- Store
- Diesel Emergency Generation Room
- Foam Package Room
- Deluge Valve Station
- Condensate Extraction Pump Building
- Valve Station for Propane and Natural Gas

4. DESIGN PARAMETER

The HVAC systems shall be designed to the following ambient climatic conditions:

- Summer: +40°C DB · +24°C WB
- Winter: -33°C DB - 95%

5. GENERAL DESIGN CRITERIA

5.1. Equipment Sparing Philosophy

The following sparing philosophy shall be provided in the design of the HVAC systems to ensure continuous operation:

- 100% stand-by units shall be provided for air handling equipment, refrigeration equipment and chilled water pumps of HVAC systems serving critical areas.
- 100% stand-by fans shall be provided for toilet exhaust air ventilation systems.

- 100% stand-by fans shall be provided for battery room exhaust air ventilation systems.

The above mentioned HVAC Systems shall be fed also from an emergency electrical supply in the event of loss of normal electrical supply.

5.2. Refrigerants

CFC type refrigerants shall not be used. HFC type refrigerants shall be provided. R-134a type refrigerant is preferred, considering the design ambient climatic conditions. The use of any other HFC type refrigerant will be permitted only upon the demonstration of the suitability for the specified ambient conditions.

5.3. Air Recirculation limitations

Air recirculation shall not be permitted from the following areas:

- Hazardous classified areas and/or areas classified for dust;
- Battery rooms;
- Toilets;
- Locker rooms;
- Janitor rooms;
- Printing rooms or photocopy rooms;
- Any other area required by ASHRAE Standard 62.1 and/or by applicable health and safety regulations.

5.4. Dedicated Exhaust Air Systems

Dedicated exhaust air ventilation systems shall be provided for the following areas:

- Battery rooms;
- Toilets, locker rooms and janitor rooms;

5.5. Clean Agent I Smoke Exhaust System

All building areas covered by Clean Agent Fire Protection Systems shall be provided with a dedicate exhaust ventilation systems, for removal of Clean Agent gas and Smoke after a fire has been extinguished. At the same time, the necessary make-up air will be supply by Air Handling Units for purging purpose only. The above mentioned systems will be designed in accordance with the requirements of Fire & Gas Detection System General Specification

5.6. Smoke Exhaust System

Smoke exhaust air system will be provided in rooms, stairs, corridors and buildings, in accordance with the requirements of Fire & Gas Detection System General Specification, Fire Protection Specification, and in accordance to local Russian and Kazakh norms, codes and standards (especially SNIP RK4.02-42-2006).

5.7. Escape Route Pressurization Systems

Pressurization System will be provided for each escape route in order to guarantee enclosed stairs or corridors free from Smoke in the event of fire.

5.8. Electrical Power Supply Requirements

5.8.1. Electrical Power Supply Characteristics

Electrical power supply for HVAC equipment shall be as follows,

- Motors from 0 kW up to 50 kW : 400 V, 3-phase, 50 Hz;
- Motors from 50 kW to 400kW: 690 V,3-phase, 50 Hz;
- Motors above 400kW: 10kV, 3-phase, 50 Hz;

5.8.2. Electrical Power Supply Incomers

HVAC systems serving non-critical areas shall be fed with a single normal power supply incomer.

HVAC systems serving areas housing vital electronic equipment that are required to operate after a normal power failure to ensure a safe plant shutdown (such as instrument rack rooms, UPS rooms, Control Room, Electrical room), shall be fed with dual incomers, one from normal power supply and one from emergency diesel generator system.

5.9. HVAC System Operations

The operations and control functions of the HVAC systems shall be fully automatic. All systems shall be designed for 24-hour per day continuous operation.

5.10. Automatic Change-Over Requirements

5.10.1. Equipment Change-Over

Change-over from duty to stand-by equipment, in case of failure of the duty unit, shall be fully automatic.

Automatic change-over shall be provided to switch, on weekly basis, the duty and the stand-by equipment, in order to ensure an even number of running hours for each unit.

5.11. SEASONAL CHANGE-OVER OF SET POINTS

Seasonal change-over in the various control set points of the HVAC systems shall be fully automatic.

5.11.1. Power Supply Change-Over

Where the HVAC systems are fed from dual power supply incomers, the HVAC motor control centers shall be provided with an automatic change-over switch for the two incomers.

5.12. ELECTRICAL HAZARDOUS AREA CLASSIFICATION

Whenever a HVAC system is located in hazardous classified area, any equipment and electrical component shall be certified for that classification. For the hazardous area classifications, refer to document Hazardous Area Classification Report and the relevant electrical documentation..

5.13. BLAST PROTECTION

In buildings subject to blast design criteria, all HVAC air duct penetrations and openings through the protected enclosure (walls, slabs or roofs) shall be provided with blast protection valves, in order to protect all the internal occupants and/or vital equipment.

5.14. FIRE PROTECTION

Generally, fire dampers shall be provided at air duct penetrations and openings through fire rated walls and/or slabs, in accordance with the requirements of NFPA 90A.

5.15. GAS PROTECTION

Motorized gas-tight dampers shall be provided at all air duct penetrations and openings through the external walls or roofs of the facilities, in order to shut-off and protect from gas infiltrations (Interface with Fire and Gas Protection Systems).

5.16. INTERCONNECTIONS WITH OTHER SYSTEMS

5.16.1. Interlocks with Fire and Gas System

To be developed by Contractor

5.16.2.Interfaces with DCS System

To be developed by Contractor

5.17. FRESH AIR INTAKES

Fresh air intakes shall be located in non-hazardous areas in order to avoid the entry from contaminated sources.

5.18. EXHAUST AIR OUTLETS

Exhaust air outlets shall be positioned as far away as possible, at least 5 m distance, from any fresh air intake in order to securely avoid any short circuiting of air between outlet and inlet. Furthermore, they shall be located in positions and at a minimum height from ground whereby they do not give rise to danger to people.

5.19. NOISE CRITERIA

Refer to SNIP 2.04-05-91, "Heating, Ventilation and Air conditioning" and to SNIP 23-03-2003 "Noise Protection".

«Para 6.3 – item 6 : Noise caused by the HVAC EQUIPMENT AND DUCTED SYSTEMS SHALL ALSO BE 10 db (A) BELOW **NOISE LEVELS STATED IN THE TABLE**. However, if appropriate, the Contractor may propose values lower than 10 db(A). If so, the contractor will submit, for Client approval, a Deviation Request (DR), with new proposed values and justifications based on assumptions concerning the noise produced by other equipment. Where necessary system shall include air duct silencers and/or external duct lining. »

5.20. HVAC SYSTEMS DESCRIPTION

5.20.1.Electrical & Distribution Substation Building

The building shall be air conditioned and pressurized by HVAC systems, as described below, in order to meet the indoor design conditions.

Cable Cellar

Ventilation system of Cable cellar shall consist of supply and exhaust fans at opposite corners of the room to maintain the required temperatures.

Clean Agent removal system will be guarantee using the same ventilation system, therefore Supply Fans, Exhaust Fans and relevant accessories will be guarantee for using at high temperature (>400°C).

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Electrical & Distribution Substation (First floor)

The HVAC system shall be all-air, single duct, constant air volume, multiple zone reheat, outdoor and return air type.

The system shall be provided with two air handling units (duty and stand-by), equipped with chilled water type cooling coil.

This building is not blast design type.

5.20.2. Control Room Building

The building, blast design type, shall be air conditioned and pressurized by HVAC systems, as described below, in order to meet the indoor design conditions.

The HVAC system shall be all-air, single duct, constant air volume, multiple zone reheat, outdoor and return air type.

The system shall be provided with two air handling units (duty and stand-by), equipped with chilled water type cooling coil.

The air handling units shall have one common outdoor air intake provided with motorized gas-tight damper and with blast valves. Preheating shall be provided at fresh air intake due to adverse ambient conditions.

One Steam humidifier shall be provided at the main supply air duct, for humidity control of the conditioned spaces.

The chilled water system shall be provided to serve the cooling coils of Control Room building AHUs. The system shall be equipped with two water-cooled chiller units (duty and stand-by) and two chilled water pumps (duty and stand-by).

Air handling units, water chiller units, chilled water pumps, fresh air intake, preheating arrangement, and humidifier shall be located in the HVAC Room.

The power/automatic control panel, to be located in the HVAC Room or in electrical room, shall be provided to feed and control all the equipment of the system.

Dedicated Clean Agent I Smoke purging ventilation systems shall be provided for DCS room false floor, Engineering room false floor, Offices, operator Control room, Telecommunication room, and server room floor protected by fire suppression gas.

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The purging ventilation system shall consist of a supply fan and relevant exhaust fan. The system shall be ducted, constant air volume, only outdoor and exhaust air type.

Pressurization System for escape route shall consist of a suitable number of supply fans in order to guarantee stairs and corridor free of smoke.

Toilet room and Lobby shall be provided with a dedicated exhaust air ventilation system with one extract fan. No air shall be re-circulated from Toilet and Lobby rooms.

Dedicated ducted zone reheat coils, electrical type, shall be provided, as necessary, for temperature control of the individual conditioned zones. The same dedicated ducted zone reheat coils shall serve purpose of maintaining desired temperature conditions during plant shutdown stage.

During plant shutdown stage, for all Air conditioned areas no separate heating system shall be provide.

5.20.3. Turbine Hall

The building shall be equipped with a ventilation system in order to control the maximum internal temperature during summer. A separate heating system will maintain the minimum internal temperature for winter season.

The Turbine Hall shall be ventilated by an all-air system, single duct, constant air volume type, 100% fresh air type, equipped with supply fans located inside the Substation's HVAC Room, at first floor plan. The fresh air will be exhausted through louvers with pressure relief dampers located on the roof, in order to guarantee the cross ventilation of the building. This system (supply fans and exhaust louvers with Pressure relief dampers) shall be used in case of fire for smoke purging purpose, therefore louvers and pressure relief dampers must be certifies as per Russian standards as smoke control devices.

During winter season, the room ventilation air changes will be reduced at minimum 6 m³/h 1m³, therefore the heating system shall be realized through a duct heating coil fed by Low Pressure Steam, installed on one supply fan only.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature " 5°C during plant shutdown.

This building is not blast design type.

5.20.4.Diesel Oil Pump House Room

The Diesel Oil Pump House Room shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam. Heating units shall be completed with fan, heating coil and outdoor air intake in order to guarantee 5 ACH.

Additional Heating Units fed by Low Pressure Steam, will be installed in order to increase the room ventilation from 5 to 8 ACH for room emergency ventilation purpose.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature $\leq 5^{\circ}\text{C}$ during plant shutdown.

Exterior walls and roof of Diesel Oil Pump house room are not blast proof design type.

5.20.5.Foam Package Room

The Foam package Room shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature $\leq 5^{\circ}\text{C}$ during plant shutdown.

Exterior walls and roof of Foam Package room are not blast proof design type.

5.20.6.Diesel Emergency Generation Room

The Diesel Emergency Generation Room shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

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Smoke exhaust and make up air system shall be provided as Kazakhstan standards.

Exterior walls and roof of Diesel Emergency generation room are not blast proof design type.

5.20.7. Clean Agent Room

The Clean Agent Room shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature $\leq 5^{\circ}\text{C}$ during plant shutdown.

Exterior walls and roof of Clean Agent room are not blast proof design type.

5.20.8.Store Room

The Store Room shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature $\leq 5^{\circ}\text{C}$ during plant shutdown.

Exterior walls and roof of Store room are not blast proof design type.

5.20.9.Deluge Valve Station

The Deluge Valve station shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature $\geq 5^{\circ}\text{C}$ during plant shutdown.

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Exterior walls and roof of Deluge Valve Station are not blast proof design type.

5.20.10. *Condensate Extraction Pump Building*

The building shall be equipped with a heating system which shall maintain the minimum internal temperature for winter season.

The heating system shall be realized through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

An additional heating system realized through a suitable number of Electrical Heating Units will be foreseen in order to guarantee in winter season the internal temperature 2: 5°C during plant shutdown.

This building is not blast design type.

5.20.11. *Valve Station for Natural Gas*

The Valve Station for Natural Gas shall be equipped with an Air Handling Unit, complete with heating coil fed by Low Pressure Steam, which shall maintain the minimum internal temperature for winter season.

The HVAC system shall be all-outdoor air, single duct, constant air volume designed for 8 ACH.

A dedicate Exhaust Fan will extract the air from the building.

The HVAC room will be heated through a suitable number of Heating Units fed by Low Pressure Steam, recirculated air type.

Exterior walls and roof of this building are not blast proof design type.